

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-a627df1d549ed45ca.jsonl`  
- SHA-256: `6f0990fee38cb940f04ba2dca2fe1c16763cd3b21faf2f278ce544a7bf2d80d6`  
- Source modified: `2026-07-02T03:01:14+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T02:59:32.850Z)

You are an adversarial verifier in a tech-debt audit. Your default stance: REFUTE. Today is 2026-07-02.

A finder claims the following about the repo at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer:

```
{  
  "title": "M4 quality-floor gate still missing ? the one unblocked item keeping the golden-fixtures track from DONE",  
  "category": "test-gap",  
  "severity": "P2",  
  "file": "docs/GOLDEN_REGRESSION_FIXTURES.md",  
  "line": 156,  
  "evidence": "GOLDEN_REGRESSION_FIXTURES.md:156 (M4 row) is ?; verified  
tests/test_golden_quality_floor.py and eval_baselines/regression_baseline.json do not exist (ls:  
No such file). $9 DoD (line 197) requires M4 before the track can flip DONE. BACKLOG.md:60 calls  
it 'recommended-next; unblocked... the cheapest step to flip the track to DONE' (advice standing  
since ~06-16).",  
  "why_it_matters": "Today the golden suite is characterization-only: it catches behavior  
CHANGES but has no committed F1 floor or windowing fingerprint, so a deliberate re-freeze after  
a SERVED retune could silently lock in worse quality. It is also the sole unblocked blocker to  
archiving the whole fixtures project.",  
  "proposed_fix": "Implement M4 per the doc's own recipe: seed per-slug floors from the F1  
already frozen in each fx_*/expected.json score (floor = current - tolerance), record a  
windowing fingerprint in eval_baselines/regression_baseline.json, keep CONTROL-only, add to the  
golden-crossos CI matrix.",  
  "effort": "M",  
  "tracked_in": "docs/GOLDEN_REGRESSION_FIXTURES.md $7 M4; docs/BACKLOG.md:60"  
}
```

Repo root candidates: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer,
C:/Users/avidu/Projects/khelsutra-guru/khelsutra, C:/Users/avidu/Projects/khelsutra-guru/sports-
data-collector, C:/Users/avidu/Projects/khelsutra-guru/rally-annotator,
C:/Users/avidu/Projects/khelsutra-guru/sports-obsreport (pick from the finding's repo field;
non-git dirs: 'C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup',
C:/Users/avidu/Projects/khelsutra-guru/khelsutra-screencast).

VERIFY against the CURRENT working tree (just pulled to origin master/main): (1) does the cited
code actually exist and behave as claimed at file:line? (2) was it already fixed by a recent
commit (git log --oneline -15 -- <file>)? (3) is the severity honest - construct the concrete
failure scenario or downgrade; (4) is the proposed fix sane and correctly scoped; (5) is it
owner-gated per: alpha/serving go-live, Studio P10/P11 corpus-promotion/training,
product/strategy/licensing, real cloud spend, counsel ToS/DPA?

HARD RULES: READ-ONLY (no file modifications, no installs, no test-suite runs). gh read-only
commands allowed. Your reasoning MUST cite file:line you actually read. CONFIRMED requires you
personally saw the defect/debt in current code; if you cannot reproduce the claim from the code,

say REFUTED; if a commit fixed it, ALREADY_FIXED.

2. user (2026-07-02T02:59:46.603Z)

```
1      # Golden Regression Fixtures ? video-free, non-attributable regression suite
2
3      > **Status:** `IN PROGRESS` ? owner-gated . **Owner:** Avi . **Created:** 2026-06-16 .
**Last updated:** 2026-06-16 . **Legal:** owner **accepted the researched risk** 2026-06-16
(conditions §3 stand; this is the owner's informed acceptance, **not** a retained-counsel
opinion)
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE, per archive policy)
5      > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
6      > **Relation to existing docs:** EXTENDS (does not supersede)
`docs/GOLDEN_DATA_SHARING.md` (#175); reuses the eval stack documented in `docs/EVALUATION.md` /
`docs/EXPERIMENT_HARNESS.md`; quality floor ties to `docs/QUALITY_ITERATIONS.md`.
7      > **Honesty note:** §3 (Legal) is **researched general information, NOT legal advice** ?
it gates on retained counsel (§8). §2/§4/§7 mark which claims are **[verified]** against code vs
**[design]** proposals.
8
9      ---
10
11     ## 0. TL;DR
12
13     Let a contributor on **another machine with no video, GPU, WSL, or DB** run the rally-
segmentation regression suite from **committed structured files**, so refactors and non-quality
PRs get a red/green side-effect signal. Two layers, built once and grown by accretion:
14
15     1. **Mechanics** ? freeze the deterministic pipeline at the **post-detector trajectory
seam** and replay it through the *already-shipped* eval stack (`experiment.py` / `metrics.py` /
`windowing.py` / `regression.py`), with two auto-discovered gates: a
**characterization/snapshot** gate (catches any behaviour change) and a **quality-floor** gate
(catches quality drops).
16     2. **Privacy/legal filter (SEALED-FIXTURES)** ? only **de-attributed, owner-source,
biometric-free** numbers are ever committed; richer/sensitive streams stay key-gated or out-of-
repo. Non-attributability comes from **deletion-at-source**, not encryption. ? "Minimized" means
**stream minimization** (shuttle-only, no person/audio/pose) ? **NOT** reduction of the shuttle
stream to ~5 scalars: the de-attributed per-frame `trajectory.csv` IS committed (see §4c + its
risk-acceptance re-confirmation note).
17
18     **The convergence that makes this clean:** the *privacy* answer (publish shuttle-only,
drop person/pose streams) and the *determinism* answer (freeze only the `CONTROL` path ? which
runs **no numpy execution**, keeping the numpy/BLAS learned scorer off the cross-machine gate)
point at the **same minimal public tier**. That coincidence is the spine of the design. (numpy
is still *imported* transitively via `experiment` and must be installed; the CONTROL replay just
never calls into it.)
19
20     **Two findings already actionable, independent of the rest:**
21     - ? **Live de-attribution bug** *(verified)*: real player/venue names are committed
today in `eval_baselines/heuristic_lovo_n6.json`,
`eval_baselines/gemini_wrapper_2026-06-10.json`, and `backend/eval/eval_partial_labels.py` (the
last also has a personal `C:/Users/avidu/...` path) ? and in git history. ? **Deliverable P0.**
22     - ? **Determinism trap** *(verified, corroborated by two independent red-teams)*: the
learned variants (`shuttle_only`, `fusion`) train a numpy/OpenBLAS `DYNAMIC_ARCH` logistic
regression whose decisions flip near the 0.5 threshold across CPUs ? would false-RED on a
contributor's machine. ? the cross-machine gate must be **CONTROL/static-only**.
23
24     ---
25
26     ## 1. Problem & goal
27
28     The rally-quality measurement is split across two machines by capability
(`docs/GOLDEN_DATA_SHARING.md`):
29     - `backend/eval/fusion_golden.py` **extracts** `*_golden_features.json` ? needs the
```

```

**video + trajectory CSV + labels + GPU/WSL** (GPU box only).
30 - `backend/eval/fusion_compare.py` / `backend/eval/ablation.py` **re-score** those JSONs
on **pure CPU, no video** ? any machine.
31
32 The CPU re-scoring is *already* video-free, but the inputs live in gitignored `output/`
/ private OneDrive, so the regression tests that depend on them **skip everywhere**
(`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`). The
committed `eval_baselines/` baselines exist but `regression.py` still scores DB-stored
predictions (needs a pipeline run on the video).
33
34 **Goal:** commit a tiny, version-tagged, **numeric-only, non-attributable** per-fixture
corpus + auto-discovered gates so a clean `git clone` / CI gets a real red/green regression
signal with **zero video/GPU/DB**, while never leaking attributable or sensitive data.
35
36 ---
37
38 ## 2. Decisions locked
39
40 | # | Decision | Source | Implication |
41 |---|-----|-----|-----|
42 | D1 | **Corpus is owner-recorded / owned** | owner answer 2026-06-16 | Legal analysis
takes **Fork A** (clean): no third-party copyright/ToS infringement; concern is trade-
secret/moat + privacy. Strongest non-attributability (no public reference to correlate against).
|
43 | D2 | **Repo private now, may open-source later** | owner answer 2026-06-16 | Design
for **public-grade hygiene from day one** ? git history is permanent; nothing scrubable-later. |
44 | D3 | **Adults, consent not formalized** | owner answer 2026-06-16 | **Person/pose/gait
streams stay OUT of the public tier** (DPDP/GDPR derived-personal-data risk without consent).
Public tier = **shuttle + abstract intervals + non-biometric features only**. |
45 | D4 | **Freeze seam = post-detector trajectory** *(design; SHIPPED differs ? see ?)* |
workflow-1 synthesis | Widest deterministic surface; reuse the shipped eval stack. ?
**Correction (audit 2026-06-16):** the shipped freeze **commits the de-attributed per-frame
(Frame,Visibility,X,Y)` `trajectory.csv`** (shuttle-only, time-origin reset) ? it is **NOT**
minimized to ~5 aggregate scalars. See §4c (and its risk-acceptance re-confirmation note). |
46 | D5 | **Cross-machine gate is CONTROL/static-only** *(design)* | both red-teams
[verified cause] | Learned variants (numpy/BLAS) are **owner-box/Tier-1 only**, asserted with
tolerance, never as a cross-machine byte-snapshot. |
47 | D6 | **Non-attributability = deletion-at-source, encryption = public-only barrier**
*(design)* | workflow-2 synthesis | A contributor who runs the tests necessarily reads
plaintext+key (DRM/analog-hole). So we **delete** attribution, not hide it. Encryption only
stops the no-key public. |
48 | D7 | **Two tiers** *(design)* | workflow-2 synthesis | **Tier-0** public/no-key
(shuttle-only, de-attributed). **Tier-1** key-gated/out-of-repo (person/audio, full corpus),
**skips cleanly when absent**. |
49
50 ---
51
52 ## 3. Legal foundation (researched ? gates on counsel, §8)
53
54 **Core verdict (settled across IN/US/EU):** the derived numbers ? per-frame shuttle
(x,y), rally start/end intervals, audio/court/aggregate-person feature values ? **do NOT inherit
the source video's copyright.** They are uncopyrightable *facts/measurements*, not authored
expression, and not a §101 "derivative work."
55
56 | Jurisdiction | Copyright of the data | Database right | Contract / ToS | Net |
57 |---|---|---|---|---|
58 | **India** (primary) | Not copyrightable as facts (*EBC v. D.B. Modak*); but
compilations get thin protection on a low "skill/labour" bar (*Burlington*, *OLX v. Padawan*) ?
only bites if **lifted from a 3rd-party dataset**, not self-derived | **None** (no sui-generis
right) ? a real permissiveness edge | Anti-scraping ToS (untested in court) + **IT Act
s.43/s.66** civil/criminal exposure for 3rd-party ingestion | **Owned-source + minors-excluded +
no person/pose = clean** |
59 | **US** | Not copyrightable (*Feist*; §102(b); *NBA v. Motorola*) | None (*Feist*
killed sweat-of-the-brow) | **Dominant risk.** ToS enforceable where copyright/CFAA fail
(*ProCD*, *hiQ* ? injunction + **forced deletion**). Remedy for redistribution is takedown, not

```

nominal damages | Publishable as facts ****iff**** owned/permissioned source + no barring ToS + no person/biometric/minor stream |

60 | ****EU / UK**** | Not copyrightable (**Football Dataco v. Yahoo**) | ****Genuinely contestable.**** **Football Dataco v. Sportradar**: recording pre-existing on-court facts = ****obtaining****, not "creating" ? our "we created it" premise is ****overconfident****; harm test = **CV-Online Latvia** | *Ryanair v. PR Aviation*: ToS can ban extraction/redistribution even with no IP right | Get EU+UK advice before redistributing numeric streams at scale |

61

62 ****Provenance is the decisive fork.**** ****Fork A** (owner-recorded, unpublished) ? the only path safe to commit publicly; collapses to a trade-secret **choice** (Indian breach-of-confidence / Contract Act 1872 / IT Act s.72 ? ****not**** US DTSA) + privacy. ****Fork B** (broadcast/YouTube/scraped) ? do ****not**** commit; the extraction act needs a license or now-****contested**** US fair use (**Thomson Reuters v. Ross** 2025; USCO May-2025), and unlawful acquisition independently sinks it (**Bartz v. Anthropic**, ~\$1.5B settlement). Caveat: owner footage ****shot at a third party's event**** (tournament/club) carries ticket/venue/organiser/broadcaster terms ? treat as Fork B until cleared.

63

64 ****Publish-by-stream (privacy):****

65 - ? ****Publish-OK:**** shuttle (x,y)+Visibility (inanimate object), abstract rally/GT ****intervals**** (identity-stripped), non-biometric audio/court/aggregate-person features detached from identity.

66 - ?? ****Conditional** (effectively never in v1): ****person bounding boxes that *persistently single out a player* ? flips to GDPR Art.9 the moment it identifies. This architecture is prone to that flip ? treat per-player tracking as a tripwire.**

67 - ? ****Never publish:**** pose/skeleton/****gait**** (re-identifies at 80?95% even with faces blurred ? **Weiss et al. 2025**), face crops, any named-player?movement mapping. Matches the existing "gait/biometrics strictly future (GDPR Art.9)" guardrail.

68 - ? ****Minors flip everything:**** DPDP "child" = under 18; s.9(3) ****prohibits behavioural monitoring**** (the Rule-12 exemptions don't cover this project). A persistent per-player signature on a minor is, conservatively, prohibited. ****Exclude minors**** from any persisted stream beyond inanimate shuttle/interval data.

69

70 ****Non-attributability is relative, not absolute.**** A per-video trajectory is a fingerprint (de Montjoye: 4 points ? 95% re-ID). It's only non-attributable because the ****Fork-A source stays unpublished**** (no public reference to correlate against). Per **EDPS v. SRB** (CJEU C-413/23 P, 2025) it ****remains personal data to the holder**** who can re-run the detector. So: conditional, revocable (publish the source and the link snaps back), forward-looking (re-assess as the public-video environment grows). Never market it as "anonymous."

71

72 ***Key authorities:** Feist 499 U.S. 340; NBA v. Motorola 105 F.3d 841; 17 U.S.C. §101/§102(b); Sega v. Accolade 977 F.2d 1510; Football Dataco v. Yahoo (C-604/10) & v. Sportradar (C-173/11); BHB v. William Hill (C-203/02); CV-Online Latvia (C-762/19); Ryanair v. PR Aviation (C-30/14); Thomson Reuters v. Ross (D. Del. 2025); EBC v. D.B. Modak; DPDP Act 2023 s.3/s.9 + Rules 2025; EDPS v. SRB (C-413/23 P). Full URLs in the research artifact (§10).

73

74 ---

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76 **## 4. Architecture (design)**

77

78 **### 4a. Freeze seam + reuse map**

79 Freeze at the ****trajectory CSV ? CONTROL-path replay****. One shared helper ``backend/eval/golden_fixtures.py::replay_fixture()`` is the **single** code path used by both the freeze tool and both gates. As ****shipped****, that path is (per the module docstring):

80 ``parse_trajectory_csv ? distill_local.build_candidates(proposal=(vt,mg,mwd)) ? VideoCandidates(5 shuttle cols only) ? experiment.predict(CONTROL) [keep all ? merge_overlaps] ? metrics.score``.

81 Reuses ****verbatim****: ``metrics.score/interval_iou`,
`experiment.predict/CONTROL/VideoCandidates`,
`distill_local.build_candidates`,
`tracknet_runner.parse_trajectory_csv`/
`trajectory_to_action_windows`,
`rally_gate` net-crossings (inside `build_candidates`),
`gt_loader.load_gt_intervals` (strict),
`regression.gt_hash`,
`eval_baselines`/
`committed-baseline precedent`.`

82 > ? ****Correction** (audit 2026-06-16): ****the replay does *NOT* import `windowing.py` ? there is no ``windowing.build_windows(preset)`` / ``WindowingPreset`` / "default SERVED" call. The windowing params travel as a ****raw 3-tuple `(vt, mg, mwd)`** pinned in each fixture's ``provenance.json`` and passed straight to ``build_candidates``; the harness deliberately avoids the**

named `windowing.SERVED` constant so a future SERVED retune can't silently churn fixtures.
 (`segmentation\_metrics`, `experiment.compare/Variant.spec\_hash`, `regression.save\_baseline`,
 `eval\_stats` are **not** on the M1 replay path either ? the quality-floor gate M4 will add the
 baseline/stats pieces.)

83

84 ### 4b. Tiered model (SEALED-FIXTURES)

85 | Tier | Where | Key? | Contents | Test behaviour |

86 |---|---|---|---|---|

87 | ****Tier-0**** | `eval\_baselines/fixtures/` (tracked) | none | opaque surrogate slug,
 fps/frame_width (retained, NOT quantized), the ****de-attributed per-frame `trajectory.csv`****
 `(Frame,Visibility,X,Y)` (shuttle-only, time-origin reset), the 5 shuttle feature columns inside
 `expected.json`, strict-slug `label`, relative `gts`. ****No person/audio/pose.**** ? NOT minimized
 to 5 scalars ? see §4c. | ****Unconditional**** ? runs on every PR/fork; turns today's skipped
 litmus into a real public gate |

88 | ****Tier-1**** | OneDrive (default) ****or**** SOPS-encrypted in-repo | token/age key | full
 corpus ****with**** person/audio; authoritative ? F1/CI/sign-test + exact Gate-A litmus | ****Skips**
 cleanly ****when key/data absent (generalize `\_golden\_present()` ? `\_tier1\_present()`)** |

89

90 A fresh no-key clone is always ****green****; coverage never depends on a secret.

91

92 ### 4c. Anti-attribution measures (Tier-0)

93

94 > ? ****RISK-ACCEPTANCE ? OWNER RE-CONFIRMED option (a), 2026-06-22.**** The shipped freeze
 (`freeze\_real\_fixture`) commits the ****de-attributed per-frame `(Frame,Visibility,X,Y)`**
 `trajectory.csv` (time-origin reset + opaque slug + metadata strip + scenario-only label) ?
****NOT**** minimized to ~5 scalars. A per-frame stream is a de Montjoye fingerprint, so non-
 attributability rests on the ****source staying unpublished**** (EDPS v. SRB), not on minimization.
 (An earlier draft wrongly claimed "5 scalars / never the per-frame CSV"; the audit corrected it
 2026-06-16.) The owner cleared ALL owned golden videos (owned + minors-free) and accepted the
 corrected per-frame basis on 2026-06-22 ? the ****6 real fixtures shipped under P3****. Owner
 alternative ? true ~5-scalar minimization ? remains available if these ever go public.

95

96 - ****Opaque RANDOM surrogate id**** (`os.urandom` + a private surrogate?source map kept
 off-git) ? ****not**** the MD5 `video\_id`, and ****not**** HMAC(salt, video_id?name) (red-team: brute-
 forceable over a tiny known roster if the salt leaks). Never persist the source `name`/filename.

97 - ****Strict-slug `label` + dropped `source\_note`**** ? the operator `label` is validated
 against `^[a-z0-9][a-z0-9\_~]\*\$` (a free-text identifier like a player/event name is refused),
 and operator free-text `source\_note` is ****accepted-but-NOT-persisted**** (never reaches the
 committed `provenance.json`) ? both close the free-text identifier leak `\_scan\_forbidden` could
 not police (audit fix #1).

98 - ****Metadata strip**** ? no filename, video_id, match/event/player identity, capture date.

99 - ****Time-origin reset**** ? subtract per-fixture `t0` (6dp-rounded) from all
 start/end/gts. ****Metric-invariant within `\_FLOAT\_TOL`**** (`metrics.interval\_iou` and start/end-
 error are pure differences): the genuine (raw) reset-vs-no-reset score delta sits at the float-
 error floor (~3.3e-7); bit-identical for whole-second offsets. See §4c-note below and §6.

100 - ****Per-frame trajectory IS committed (de-attributed), NOT minimized to scalars**** ?
 Tier-0 commits the de-attributed per-frame `trajectory.csv` (shuttle stream only). It is ****not****
 reduced to ~5 aggregate scalars; the 5 shuttle ****feature**** columns appear (computed) inside
 `expected.json`, but the raw per-frame `(Frame,Visibility,X,Y)` stream is part of the committed
 fixture. (This is the corrected basis the ? note above gates on.)

101 - ****Stream minimization (streams, not resolution)**** ? Tier-0 keeps the shuttle block
 only; person/audio excluded; pose/gait/face have ****no field in the schema by design**** (positive
 `\_scan\_forbidden` assertion on the written bytes).

102 - ****fps/frame_width are RETAINED as scale constants**** ? they are NOT quantized today
 (quantization is a deferred P3 option, see §7 ?). Reconcile any "quantize fps/frame_width"
 wording elsewhere to this.

103 - ****Provenance hard-fail gate**** ? the de-attribution pass refuses to emit a public
 fixture for any record not tagged owner-recorded(Fork-A) + minors-excluded + non-blank-license,
 and refuses a negative reset label (timeline predates the trajectory). Writes go to a temp dir
 and are promoted only after the privacy scan passes, so a refused freeze leaves no partial
 fixture (audit fixes #3a/#3b).

104

105 ### 4d. The two gates

106 - ****Characterization/snapshot**** ? replay fixture, ****parse-compare**** JSON (never byte-
 compare ? CRLF/float-format safe; `core.autocrlf=true` here), exact on int/structural fields,

```

`approx(abs=1e-6)` on floats; stamp + check windowing-preset/label/gt fingerprints
(incommensurable ? loud fail). Catches any behaviour change even at constant F1. **CONTROL path
only** across machines.
107 - **Quality-floor** ? score vs committed labels; assert per-video + mean F1 ? floor in
`eval_baselines/regression_baseline.json` (created here; carries a **windowing fingerprint** so
a served-default change is flagged incommensurable, not silently compared); emit `eval_stats`
paired-delta-CI + sign test (NUMBERS INFORM); add a **metrics-vs-base-branch** paired delta.
Synthetic-tier floors measure **correctness, not field quality** (documented).
108
109     ### 4e. Serialization & optional protobuf
110     JSON with `sort_keys=True` + pre-quantized floats + LF is sufficient. **Protobuf is
optional (01)** ? adopt only if cross-OS byte-stable snapshots are wanted (proto3
`rally.fixture.v1`, codegen + `protoc && git diff --exit-code` guard, `*.pb binary -text`).
Protobuf is **encoding, not secrecy** (trivially `--decode_raw`).
111
112     ### 4f. Encryption & keys (Tier-1 only)
113     **Default = out-of-repo + token** (existing OneDrive `RALLY_GOLDEN_DIR` seam ? zero
sensitive bytes in git history, sidesteps the legal "redistribution" act). **Alternative = SOPS
+ age** (per-recipient, revocable, value-level diffs) for Fork-A-clean-but-private data wanting
in-repo versioning. Rejected: git-crypt (whole-repo re-key), git-lfs (storage ? secrecy). CI
secret exposed **only to the Tier-1 job on trusted branches, never to fork-PR runners**. The
**HMAC/surrogate salt** (if used) and the surrogate?source map live with the private corpus,
rotated per release. Document explicitly that Tier-1 crypto defends **only** the no-key public.
114
115     ---
116
117     ## 5. Threat model (who's protected vs not)
118
119     | Actor | Protected against | NOT protected (by design / residual) |
120     |---|---|---|
121     | **TM1 no-key public** | map fixture?source/match/player; decrypt Tier-1; confirm a
source even holding the candidate file | read Tier-0 numbers + run shuttle-only harness
(intended; useless-in-isolation) |
122     | **TM2 keyed contributor** | recover filename/video_id/match/face/pose/gait ? *never
written* (survives the analog hole) | read every value the suite reads (unpreventable; not
pretended otherwise) |
123     | **TM3 CI runner** | leaked log can't leak attribution it never had | runs the suite
(intended); Tier-1 secret never on fork runners |
124     | **TM4 malicious keyed insider** | correlation needs a public reference; Fork-A source
is unpublished ? "not reasonably likely" | **residual:** conditional + revocable ? publish/leak
the source later and the link can snap back (forward-looking re-test duty) |
125     | **TM5 re-ID adversary** | source unpublished + reset timeline ? no back-match against
a public reference | ? becomes live if a Fork-A source is ever published ? **and the committed
per-frame trajectory IS a fingerprint** (de Montjoye 4-point re-ID), NOT just ~5 scalars
(corrected §4c). The protection is the unpublished source, not minimization |
126     | **TM6 accidental plaintext commit** | required CI leak-scan rejects MD5/known-
name/non-surrogate/plaintext-SOPS | if the required check is disabled, a leak lands |
127
128     ---
129
130     ## 6. Honest limits ? what a green suite does NOT prove
131
132     - Green proves only that the **deterministic post-trajectory transforms** are unchanged
for the **frozen cases**. It does **not** cover: the WASB/TrackNet detector,
`_probe_fps_width`/cv2 frame-peek, the AI/provider handoff, chunking/re-encode, the ffmpeg
stitch, DB persistence/migrations, or the API filter. *A refactor that breaks decoding or the
detector passes these gates green.* The owner-box `regression.py` DB run remains the only full-
chain check.
133     - Characterization = **behaviour-locking, not correctness** ? it freezes current
behaviour *including current bugs*. "Passing" = "behaviour unchanged", never "behaviour
correct". This caveat must ride in the **test pass/fail message**, not only the docs.
134     - **Synthetic-tier floors = plumbing correctness, NOT field quality.** Never cite
synthetic F1 in `QUALITY_ITERATIONS.md` / the paper as rally-detection performance (tag `kind:
synthetic`).
135     - Tier-0 public coverage is **shuttle-only**; the person/audio fusion ?F1 and exact

```

Gate-A values need Tier-1 (skipped on no-key clones).

136 - Data is **not** "anonymous" absolutely (EDPS v. SRB) ? relative, conditional, revocable, forward-looking.

137 - The committed Tier-0 stream is the **de-attributed** per-frame `trajectory.csv`, NOT ~5 scalars ? a per-frame trajectory is a de Montjoye fingerprint; the only protection is the **unpublished source** (§4c ?, §5 TM5).

138 - **Time-origin** reset is metric-invariant within `_FLOAT_TOL`, not byte-exact for every offset. The genuine (raw) reset-vs-no-reset score delta is at the float-error floor (~3.3e-7); it is **bit-identical** for whole-second offsets, but because 6dp-rounding is not additive (`round6(a)-round6(b) ? round6(a-b)`) the **serialized** 6dp boundary error can differ by one 6dp ULP (1e-6) on a sub-second offset ? the serialization grid, not a metric divergence. (Audit fix #2 rounds `t0`+labels to 6dp before the shift to keep the residual ? the grid step.)

139 - Encryption gives **no** secrecy from a running contributor; protobuf gives **no** secrecy at all.

140 - De-attribution does **not** cure provenance/licensing ? it sits **behind** the per-record provenance gate + counsel sign-off.

141 - `--refresh-snapshot` can rubber-stamp a real regression; mitigation is **review** of the visible diff + a separate, deliberate floor-baseline update. **A** recorded `--reason` is **REQUIRED** (the CLI errors without one ? audit fix #6).

142

143 ---

144

145 **## 7. Deliverables & progress tracker ? source of truth**

146

147 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One **small PR per row**, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.

148

ID	Deliverable	Depends on	Counsel-gated?	Status	PR
149	149 ID Deliverable Depends on Counsel-gated? Status PR				
150	150 --- ----- ----- ----- ----- ----				
151	151 P0 PII leak hygiene (partial ? paths + scan). Scrubbed personal <code>C:\Users\?</code> paths from code docstrings (<code>eval_partial_labels.py</code> , <code>gemini_refine.py</code> , <code>README.md</code>) + added <code>tools/leak_scan.py</code> (32-hex MD5 + personal-path structural guard, optional gitignored <code>.leakscan_private</code> name-list) + a CI <code>leak-scan</code> job + tests. ? The functional video-name rename (<code>eval_baselines/*.json</code> keys / <code>rally_seg_eval.MULTICOURT_CLIPS</code> / quality-log names) is DEFERRED per the owner's safe-scope decision (2026-06-16) ? it's coupled to the gitignored <code>output/human_lovo_manifest.json</code> + data files + the training floor and collides with the active quality track; run the private-list scan + a coordinated rename before open-sourcing. Git-history rewrite (PR #77) also deferred. ? No ? <code>_this PR</code>				
152	152 M1 Fixture framework + synthetic Tier-0 + shared replay helper (also delivered M2-core + M3 ? see below). <code>backend/eval/golden_fixtures.py</code> (<code>replay_fixture</code> , schema dispatch, <code>Path(__file__)-relative</code>) + 3 tool-generated synthetic fixtures + manifest + <code>README</code> + <code>.gitattributes</code> LF pin + <code>tests/test_golden_fixtures.py</code> . CONTROL-only (no numpy execution on the replay path); cross-OS verified. ? No (synthetic) ?				
153	153 M2 Freeze/refresh extractor . Core (<code>freeze_synthetic</code> / <code>refresh_snapshot</code> / <code>--check</code> CLI + idempotence test) shipped in M1 (#189); the <code>--license</code> / <code>--minors-free</code> / <code>--owned</code> refusal gate shipped with P1 (this PR). M1 No ?				
154	154 M3 Characterization gate ? <code>tests/test_golden_fixtures.py</code> : parse-compared (exact-int / approx-float 1e-6), CONTROL-only , positive no-person/audio assertion, perturbation + idempotence. Shipped in M1 (#189) . (Windowing/label fingerprint guard + row-count cap ? fold into M4.) M1 No ?				
155	155 MX Cross-OS determinism guard (CI) . Lightweight <code>golden-crossos</code> CI job (ubuntu/windows/macos) running <code>--check</code> + the fixtures test ? locks the cross-OS guarantee (CONTROL replay is numpy-only, no GPU stack; a separate fast job , not full-suite×3). M1 No ?				
156	156 M4 Quality-floor gate + first regression baseline . <code>tests/test_golden_quality_floor.py</code> ; create <code>eval_baselines/regression_baseline.json</code> with windowing fingerprint ; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for runtime. M1, M2 No ? ?				
157	157 P1 De-attribution + minimization ? <code>golden_fixtures.freeze_real_fixture()</code> (extends the M1 module, not a separate script): opaque random <code>fx_hex</code> surrogate slug, metadata strip, metric-invariant time-origin reset , strict GT loader, positive no-person/audio/MD5/source-path assertion , manifest non-clobber; <code>--freeze-real</code> CLI with the				

```

**refusal gate** (exit 2 unless owned + minors-free + license). Tooling-only, 12 synthetic
tests; **no real data committed**. | M1 | No (tooling) | ? | _this PR_ |
158 | **P2** | **Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the
researched IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source
minors excluded · biometric/person streams excluded · de-attributed). Owner's informed risk
acceptance ? **NOT a retained-counsel opinion**; optional future counsel confirmation per §8. |
P1 | Owner-accepted | ? | ? |
159 | **P3** | **Tier-0 REAL subset ? SHIPPED.** Committed **6 curated, trimmed (?12k-frame
? few-min), de-attributed** real fixtures (`fx_r01`?`fx_r06`: single-court / multi-court×2 /
doubles×2 / continuously-active blob; both 29.97 & 59.94 fps) via `freeze_real_fixture --max-
frames` (opaque slug, time-origin reset, scenario-only labels, **NO** source name/path/video_id
? grep-verified + leak-scan clean). ~1.9 MB total; the characterization + cross-OS gates now run
on real data too. Owner cleared ALL owned golden videos (owned + minors-free) + re-confirmed
option (a) (2026-06-22). The surrogate?source map stays **off-git** (owner-held). | P1, P2 |
conditions §3 ? | ? | _this PR_ |
160 | **O1** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1`
proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
161 | **O2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
162 | **DOC** | **Discoverability cross-links** ? added the `eval_baselines/fixtures/` row
to `CODE_MAP §0` + a `golden_fixtures.py` entry to `CODE_MAP §2.3`; cross-linked
`GOLDEN_DATA_SHARING.md` (committed subset ? private OneDrive corpus). README indexed in #186;
memory `current-state` kept live. (EVALUATION/QUALITY_ITERATIONS cross-links **deferred** ? that
doc surface is hot in the concurrent quality track; avoid collision.) | M1 | No | ? | _this PR_
|
163
164 **Must-fix-before-any-public-commit (red-team), folded into the rows above:** ? numeric-
tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5/MX ?); ? scrub names in
**all** files + history decision (P0); ? leak-scan a **required** check, hooks are convenience
only (P0); ? person-optional loader + **positive** no-person assertion, don't trust the silent
`0.0` default (P1 ?); ? random surrogate id not HMAC-over-roster (P1 ?); ? quantize/drop
fps/frame_width (P1 ? fps/fw retained as scale constants; quantize is a P3 option); ? document
Tier-0 omits the fusion path (DOC); ? counsel + Fork-A + minors/biometric exclusion as hard
preconditions (P2 owner-accepted; enforced in code by P1's refusal gate ?).
165
166 ### Progress log (quantitative ? coverage must hold/improve, CI incl. the 3-OS cross-OS
job green)
167 | PR | Deliverable(s) | Suite | Coverage | Notes |
168 |----|-----|-----|-----|-----|
169 | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) | M1 +
M2-core + M3 | 746 pass / 7 skip | 80.41% | synthetic Tier-0 + CONTROL replay +
characterization; cross-OS proven (CI Linux replayed Windows-frozen fixtures) |
170 | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) | MX +
tracker/legal reconcile | full suite green | 80.41% | `golden-crossos` CI job green on
ubuntu+windows+macos ? cross-OS **enforced**; legal ? owner-accepted |
171 | _this PR_ | P1 + M2 refusal gate | 759 pass / 7 skip | **80.49%** (+0.08) |
`freeze_real_fixture` + refusal gate; **+13 tests**; tooling-only, no real data committed |
172 | _this PR (audit-hardening)_ | adversarial-audit fixes #1?#17 | 784 pass / 7 skip |
**80.71%** (+0.22) | strict-slug `label` + dropped `source_note` (#1); 6dp-rounded reset within
`_FLOAT_TOL` (#2); temp-dir-then-promote + negative-label refusal (#3); exact-membership source-
path scan (#4); key-set-equality in `compare_expected` (#5); `--reason` mandatory (#6);
independent schema guard (#7); `max_win` recorded-not-applied note (#8); numpy-execution wording
(#9); +failure-semantics & by-kind tests (#10/#11); `test_golden_real_fixtures.py` added to
`golden-crossos` CI (#12); doc reconciled to shipped reality incl. per-frame-CSV honesty + risk
re-confirm (#13?#17). Synthetic `--check` byte-identical (unchanged). |
173 | _this PR (P0 safe scope)_ | personal-path scrub + leak-scan guard | 1365 pass / 7 skip
| 85.27% | scrubbed `C:\Users\?` paths from 3 code docstrings (`eval_partial_labels.py` /
`gemini_refine.py` / `README.md`); `tools/leak_scan.py` (32-hex MD5 + personal-path structural
guard + gitignored `.leakscan_private` name-list) + a CI `leak-scan` job + 10 tests. Functional
video-name rename DEFERRED (owner safe-scope). |
174 | _this PR (P3 real fixtures)_ | 6 curated trimmed de-attributed real Tier-0 fixtures +
`--max-frames` trim | 1390 pass / 7 skip | 85.27% | `fx_r01`?`fx_r06` (~1.9 MB; single/multi-
court/doubles/blob, both fps) frozen via `freeze_real_fixture --max-frames=12000`; de-

```



```

attribution grep-verified + leak-scan clean; +trim test; owner re-confirmed option (a). F1
0.0?0.59 = real CONTROL-path (behaviour-locking, **NOT** field quality). |
175
176 ---
177
178 ## 8. Open owner / counsel questions
179
180 **Counsel questions ? documented basis (the owner accepted the researched risk on
2026-06-16; see P2). These remain the record, and a retained-counsel confirmation is
optional/recommended before scaling beyond a small owned, minors-free, de-attributed subset:**
181 1. Does the EU/UK **sui-generis database right** subsist in our self-derived
shuttle/rally streams on the *Sportradar* "obtained vs created" reasoning, and does
redistributing a de-attributed Tier-0 subset cause *CV-Online* harm?
182 2. Post *Thomson Reuters v. Ross* + USCO-2025, is **US fair use for the extraction
step** defensible if the dataset could compete in a "data/analytics" market ? does publishing
only numbers (never footage) matter?
183 3. India: are anti-scraping ToS enforceable; **IT Act s.43/s.66** exposure for any 3rd-
party ingestion; does pending *ANI v. OpenAI* (~Apr 2026) move it? Confirm **DPDP s.9/Rule 10**
and whether per-player tracking is prohibited s.9(3) monitoring of a minor.
184 4. Sign-off on the **final committed Tier-0 subset + provenance log** ; is non-
attributability-via-private-source a defensible posture to assert given its
conditional/revocable nature?
185
186 **Owner (strategic):**
187 5. Publishing the Fork-A subset **forfeits the trade-secret moat** (Indian breach-of-
confidence framing) ? weighed against community/benchmark/paper aims. Your call.
188 6. **Forward-looking commitment:** do not publish a highlight reel of the *same* source
whose fixtures are committed (without re-clearing), and accept a periodic re-assessment as the
public-video environment grows.
189 7. **Scope of first public commit:** synthetic-only (zero provenance risk, recommended)
vs a counsel-cleared real subset.
190 8. **Tier-1 mechanism:** out-of-repo+token (recommended) vs SOPS+age in-repo. And: is
protobuf (01) wanted, or is canonical-JSON enough?
191
192 ---
193
194 ## 9. Definition of done
195
196 The project is **DONE** (flip `Status ? DONE`, archive) when:
197 - [ ] **P0, M1?M4, P1, DOC** are ? and merged.
198 - [ ] A clean `git clone` on **win + ubuntu + mac** runs `python run_all_tests.py` and
the **Tier-0 gates pass green with no key / no video / no GPU / no DB**.
199 - [ ] The **CI leak-scan is a required status check** and passes (no MD5 / real-name /
non-surrogate / plaintext-SOPS in the tree).
200 - [ ] Characterization + quality-floor gates are **parse-compared, CONTROL-only,
windowing-fingerprinted**, and demonstrably catch a seeded behaviour change.
201 - [ ] `docs/CODE_MAP.md §0` + TEST MAP updated; this doc cross-linked; `NEXT_STEPS.md` +
memory `current-state` point here.
202 - [ ] **Either** the public commit is synthetic-only, **or** (**P2 owner-accepted ? +
P3**) for a real-data commit, under the §3 conditions enforced **in code** by P1's refusal gate.
203 - [ ] §6 honest-limits caveat rides in the test pass/fail messages, not only the docs.
204
205 *Optional 01/02 may remain ? without blocking DONE (they're earned, not assumed).*
206
207 ---
208
209 ## 10. References
210
211 **Internal code anchors (verified):** `backend/eval/{regression.py, fusion_golden.py, fusi
on_compare.py, ablation.py, experiment.py, metrics.py, windowing.py, distill_local.py, gt_loader.py, ev
al_stats.py}` · `backend/pipeline/detectors/{base.py, tracknet_runner.py, rally_gate.py}` ·
`backend/utils/hashing.py::compute_video_id` · `eval_baselines/{heuristic_lovo_n6.json, gemini_wr
apper_2026-06-10.json, experiment_leaderboard.json}` ·
`tests/test_ablation.py::test_litmus_reproduces_gate_a` · `.github/workflows/ci.yml`.
212 **Internal docs:** `GOLDEN_DATA_SHARING.md` (#175), `EVALUATION.md`,

```

```
`EXPERIMENT_HARNESS.md`, `QUALITY_ITERATIONS.md`, `MULTI_SIGNAL_FUSION_PLAN.md`, `CODE_MAP.md`  
S0`.  
213     **Legal authorities:** see §3 (full URLs in the research workflow artifact). *This  
section is researched general information, not legal advice.*  
214  
215     ---  
216  
217     ### Changelog  
218     - **2026-06-16** ? Initial draft synthesized from two multi-agent workflows (mechanics  
design + 2 critiques; legal memo IN/US/EU + SEALED-FIXTURES architecture + red-team). Verified  
the live name-leak and the BLAS-determinism cause against code. Status: DRAFT, owner-gated.  
219
```

3. user (2026-07-02T02:59:47.465Z)

No files found

4. user (2026-07-02T02:59:47.539Z)

No files found

5. user (2026-07-02T02:59:57.850Z)

```
experiment_leaderboard.json  
fixtures  
gemini_wrapper_2026-06-10.json  
heuristic_lovo_n6.json  
nightly_baseline.json  
reelcount_manifest.json  
test_fusion_golden_telemetry.py  
test_golden_fixtures.py  
test_golden_fixtures_split.py  
test_golden_real_fixtures.py
```

6. user (2026-07-02T02:59:59.197Z)

```
badminton-highlight-indexer\docs\BACKLOG.md:60:[Omitted long matching line]  
badminton-highlight-indexer\backend\eval\experiment.py:31: - the promotion gate stays  
`run_regression.py` + `regression_baseline.json`  
badminton-highlight-indexer\backend\eval\experiment.py:685:     `regression_baseline.json`;  
deliberately the size of a baseline file, not a service  
badminton-highlight-indexer\backend\eval\per_user_regression.py:4:``regression_baseline.json``):  
a candidate is gated so a global change can never silently  
badminton-highlight-indexer\backend\eval\per_user_regression.py:37:saved/loaded as JSON, a  
sibling to `regression.py`'s ``regression_baseline.json`` approach.  
badminton-highlight-indexer\backend\eval\regression.py:34:DEFAULT_BASELINE_PATH =  
os.path.join("eval_baselines", "regression_baseline.json")  
badminton-highlight-indexer\docs\GOLDEN_REGRESSION_FIXTURES.md:107:- **Quality-floor** ? score  
vs committed labels; assert per-video + mean F1 ? floor in  
`eval_baselines/regression_baseline.json` (created here; carries a **windowing fingerprint** so  
a served-default change is flagged incommensurable, not silently compared); emit `eval_stats`  
paired-delta-CI + sign test (NUMBERS INFORM); add a **metrics-vs-base-branch** paired delta.  
Synthetic-tier floors measure **correctness, not field quality** (documented).  
badminton-highlight-indexer\docs\GOLDEN_REGRESSION_FIXTURES.md:156:| **M4** | **Quality-floor  
gate + first `regression_baseline.json`.** `tests/test_golden_quality_floor.py`; create  
`eval_baselines/regression_baseline.json` with **windowing fingerprint**;; metrics-vs-base  
paired-delta; corpus-level (not per-slug) LOVO for runtime. | M1, M2 | No | ? | ? |  
badminton-highlight-indexer\docs\EXPERIMENT_HARNESS.md:81:- Default baseline file:  
`eval_baselines/regression_baseline.json` (:33).  
badminton-highlight-indexer\docs\CODE_MAP.md:182:[Omitted long matching line]  
badminton-highlight-indexer\tests\test_per_user_regression.py:157:     """A user_id::stage  
baseline record survives save ? load unchanged (sibling to regression_baseline.json)."""
```

7. user (2026-07-02T03:00:00.045Z)

40 - ? ****Real-video validation of the swap**** ? the torchvision + ByteTrack Stage-1 is unit-tested with mocks but ****not yet validated on actual footage**** for detection/tracking quality. (ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so expect tuning.) ****Revisit:**** when a real test clip + the pipeline are run together.

41 - ? ****`yolo\_hybrid` naming**** ? the registry key, `yolo\_` config keys, and `yolo-hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining. Optional rename. ****Revisit:**** if/when a clean break is worth the migration.

42

43 **## Golden-set subproject**

44 ***(source: `docs/GOLDEN\_SET\_IMPLEMENTATION.md`, `docs/GOLDEN\_SET\_VENDORING.md`)***

45

46 - ? ****Annotation guideline**** ? ****DONE (2026-06-02):**** authored as the Roora brief + illustrated guide in the collector repo `docs/annotation/` ? rally CSV schema; ignore dead time / include every rally (incl. faults & lets) / shot = one contact; `ending\_reason` (forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for badminton & pickleball.

47 - ? ****Internal gold key**** ? hand-annotate 5?10 licensed clips to near-perfect as the vendor answer key. ****Unblocked.**** ****Revisit:**** anytime.

48 - ? ****GS-3 ? `HumanVendorProvider`**** ? concrete adapter for the chosen annotation vendor (submit/poll/fetch + DPA/secure-link controls). ****Vendor selected for the initial pilot: Roora**** (Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). ****Revisit:**** once the Roora pilot confirms quality, then build the adapter.

49 - ? ****GS-4 oracle model**** ? `AutomatedProvider` interface is stubbed (this work) with a deterministic fake. Wire a ***concrete*** fast-tier auto-labeler later. ****Oracle rule:**** it **MUST** be a different model lineage than production, or the regression test is blind to shared failures. ****Revisit:**** when a production LLM step is chosen (pick the oracle's lineage to differ).

50 - ? ****GS-5 ? regression triggers**** ? wire `regress()` into CI (pre-merge gate), on-demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path. ****Revisit:**** after GS-4.

51 - ? ****Vendor pilot (Roora)**** ? run the paid pilot with ****Roora**** as a ****single-vendor pilot first**** (not a multi-vendor bake-off): assess quality-per-rupee against the \$4 bar; escalate to the iMerit/Taskmonk fallback only if it misses. ****Revisit:**** the pilot doubles as calibration once the gold key exists.

52 - ? ****Phase-1 video collection**** ? gather the round-1 golden corpus per `docs/GOLDEN\_SET\_PHASE1\_VIDEOS.md`: ****static self-recorded**** footage (mirror deployment, not broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with ****? held out by match****. ****Unblocked.****

53 - ? ****Train/eval separation guardrails**** (was `DEFERRED\_HARDENING\_PLAN` #13; design preserved here on archival 2026-06-14) ? train & eval videos live in ****separate catalogs**** today, kept apart by convention only; nothing ***enforces*** that a video used to train the learned segmenter (`backend/eval/training.py::build\_training\_set` ? `distill\_local`) is absent from the eval/regression set, so leakage would silently inflate the headline metrics. ****Concrete design**** (from the archived DEFERRED plan \$2): split unit = ****match, never clip**** (hold ~? of each sport as eval, per `GOLDEN\_SET\_PHASE1\_VIDEOS.md`); add a `split: "train"|"eval"` field to the collector's `curate export` manifest (cross-repo contract, ADR-002); ****[DONE ? pure-logic core, PR #171]**** `backend/eval/splits.py` now exists with `load\_split(manifest) -> (train\_ids, eval\_ids)` keyed by ****match id**** (not file MD5 ? re-encodes share a match but differ in `compute\_video\_id`) + `assert\_disjoint(train\_ids, eval\_ids)` that raises on overlap, with `tests/test\_splits.py` (overlapping ids ? `assert\_disjoint` raises). ****STILL DEFERRED:**** wire the guard into the training entrypoint (`distill\_local`/`training`) and into `run\_regression.py` (baseline computed strictly on the eval split); add the collector's `curate export` manifest `split` field (above); the integration test (a deliberately-leaked manifest fails the training entrypoint). Full rationale/risks: archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md \$2. ****Revisit:**** dedicated design session with the user (needs a collector-side manifest change).

54

55 **## Golden regression fixtures (video-free, non-attributable regression suite)**

56 ***(source: `docs/GOLDEN\_REGRESSION\_FIXTURES.md` \$7 tracker / \$9 definition-of-done)***

57

58 **Core suite ****shipped**** (M1·M2·M3·MX·P1·P2·P3·DOC ? ? synthetic + 6 de-attributed real `fx\_r01`?`fx\_r06` fixtures replay cross-OS, leak-guarded, CONTROL-only). The two items below are the ****gating remainder**** of the doc's \$9 definition-of-done (it flips to `DONE`/archive only when ****P0 + M1?M4 + P1 + DOC**** are ?); 01/02 are explicitly optional. Tracked here so the track isn't silently treated as "done" while DoD items remain.**

59

```

60      - ? **M4 ? quality-floor gate + first `regression_baseline.json`** *(recommended-next;
unblocked).* Assert per-fixture **and** mean F1 ? a committed floor
(`tests/test_golden_quality_floor.py` + `eval_baselines/regression_baseline.json`). *First
move:* seed the per-slug floor from the F1 already frozen in each `fx_*/expected.json` `score`
(real fixtures sit at **0.0?0.59**, so floor = `current ? tolerance`, not a global bar).
**Gotchas:** record a **windowing fingerprint** in the baseline (else a SERVED retune silently
passes); keep it **CONTROL-only** (learned/numpy-BLAS is non-deterministic across CPUs); add it
to the `golden-crossos` CI matrix. **Revisit:** anytime ? it's the cheapest step to flip the
track to DONE.
61      - ? **P0 remainder ? functional video-name rename** *(deferred per owner safe-scope,
2026-06-16).* Rename the functional names (`eval_baselines/*.json` keys,
`rally_seg_eval.MULTICOURT_CLIPS`, quality-log names) ? opaque ids. *Why it's hard:* the keys
feed `training/gen0/harness.py`'s LOVO floor matched to the **gitignored
`output/human_lovo_manifest.json` + data files** ? rename them together or the floor breaks;
also collides with the hot quality track. *First move:* put the names in a gitignored
`.leakscan_private` and run `python -m tools.leak_scan --names .leakscan_private` to enumerate
every occurrence, then plan the coordinated rename. **Revisit:** before any open-sourcing /
public commit.
62      - ? **01 ? protobuf + deterministic serialization** *(optional).* `rally.fixture.v1`
proto3 + codegen + `protoc`/`git diff --exit-code` guard. **Revisit:** only if cross-OS **byte-
stable** snapshots are wanted (JSON parse-compare suffices today).
63      - ? **02 ? Tier-1 key-gated path** *(optional).* `scripts/fetch_golden.py` (OneDrive
token) and/or SOPS+age, `RALLY_GOLDEN_DIR` wired, Tier-1 CI gated on a secret (never on fork
runners). **Revisit:** only if sharing person/biometric streams (Tier-1) becomes necessary.
64
65      ## Recording best practices
66      *(source: `docs/VIDEO_RECORDING_GUIDELINES.md`)*
67
68      - ? **Validate recording hypotheses** ? empirically test which capture conditions
(resolution, fps, camera angle/height, lighting) actually move precision/recall, then promote
the winners into user-facing "how to record" guidance. Only the static-camera rule is currently
evidence-backed (ADR-004). **Revisit:** once a controlled corpus + the eval loop can measure
per-condition deltas.
69
70      ## Code & test organization
71      *(source: owner review comments on PR #35, 2026-06-07)*
72
73      - ? **`tests/` reorganization to reduce fragmentation** ? owner wants to isolate test
components on a grouping
74      metric (by subsystem/component) so `tests/` is less fragmented, mirroring the new
`backend/pipeline/audio/` +
75      `backend/eval/audio/` layout (e.g. audio tests grouped under `tests/audio/`, eval
tests under `tests/eval/`).
76      The exact grouping **metric is the owner's design choice** (consolidate small files
vs. add per-area dirs ? the
77      "lesser fragmentation" goal slightly tensions with adding dirs), so this is left as a
**dedicated refactor PR**,
78      not guessed here. **Revisit:** when the owner specifies the grouping scheme. (The
`backend/eval/audio/` half of
79      the request ? comment 1 ? was done in PR #35.)

```

8. user (2026-07-02T03:00:19.774Z)

```

7a40434 feat(eval): golden fixtures P3 ? 6 curated, trimmed, de-attributed REAL Tier-0 fixtures
+ --max-frames trim (#317)
5c2eec8 chore(security): P0 (safe scope) ? scrub personal paths from code + add leak-scan CI
guard (#314)
7d84319 fix(eval): golden-fixtures audit hardening ? close source_note/label leak + 16 confirmed
audit fixes (#194)
085c07b docs(eval): golden-fixtures discoverability ? CODE_MAP §0/§2.3 + GOLDEN_DATA_SHARING
cross-link (DOC) (#193)
8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
(#192)
adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)

```

```

1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)
---
d301a7f fix(nightly): GCP-L4-validated fixes ? gs:// fetch dst, WASB env, weights path, pure-
engine (#318)
fa2342c feat(nightly): E2E engine reel-count regression on a GCP GPU VM (emailed) + cost (#315)
7a40434 feat(eval): golden fixtures P3 ? 6 curated, trimmed, de-attributed REAL Tier-0 fixtures
+ --max-frames trim (#317)
eff12c0 chore(eval): re-baseline the nightly regression to the n=15 golden corpus (#239)
a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)
ff51b37 chore(eval): re-baseline the nightly after the R1 milestone flip (#204) (#216)
2eaccd5 feat(eval): nightly rally-quality regression tripwire (golden corpus, CPU) (#202)
7d84319 fix(eval): golden-fixtures audit hardening ? close source_note/label leak + 16 confirmed
audit fixes (#194)
---
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)
aa915a7 deploy/gcp: centralize project + bucket names in one sourced buckets.env (#410)
52d97b5 P4: POST /api/assets ? store a video into the user's chosen medium (#409)
---

```

master...origin/master

9. user (2026-07-02T03:00:20.780Z)

```

{
  "_doc": "Frozen nightly rally-quality regression baseline. Regenerate with `python
scripts/nightly_regression.py --update-baseline` after an intended improvement or a corpus
change. Tied to the LOCAL golden corpus (not committed).",
  "configs": {
    "baseline_off": {
      "aggregate": {
        "dend_abs_median": 41.77921047714383,
        "dstart_abs_median": 42.467243955066174,
        "f1@0.25": 0.39981006941991765,
        "f1@0.5": 0.2366145803119009,
        "merge_rate": 0.6380248744753217,
        "precision": 0.2577557018251538,
        "recall": 0.22719343607869802,
        "segment_count_ratio": 0.5824947574930804,
        "split_rate": 0.0,
        "tol_f1": 0.1382064040388172,
        "tol_precision": 0.14536686157033415,
        "tol_recall": 0.13445461
      }
    }
  }
}

```

```

README.md
clean_single_rally
fx_r01
fx_r02
fx_r03
fx_r04
fx_r05
fx_r06
manifest.json
multi_rally_with_deadtime

```

10. assistant (2026-07-02T03:00:57.772Z)

All claims verified against the current working tree. Verdict is CONFIRMED.

11. user (2026-07-02T03:01:14.623Z)

Structured output provided successfully